Health and Safety. OSH Act, OSHA and requirements for Architecture & Water Engineering U.T.C Elgon 2021

OCCUPATIONAL HEALTH SAFETY

Safety *means preventing accidents*. **Health** *is being free from any form of illness in the body and mind.*

International Labour Organization (ILO) and (WHO) *defined occupational health as the promotion and maintenance of the highest degree of physical and social wellbeing of workers in all occupations.*

History of Uganda Safety and Health Legislation

In Uganda, a Factories Inspectorate was launched as far back as 1925 within the Department of Labour. From 1948 to 1959, the Department of Health seconded a senior Medical Officer to the Department to look after the health of the workers. In 1965, a full-time, fully trained Ugandan Occupational Health Physician started an Occupational Health Unit in the Ministry of Labour, which was separate from the Factories Inspectorate. This Unit grew and was later renamed the Occupational Health and Hygiene Department. The Department of Factories had meanwhile also grown and expanded to establish two up-country stations in the two other most industrialized towns of Uganda, i.e. Jinja and Mbale. In 1984 and 1985, the International Labour Office (ILO) fielded and funded two missions to Uganda. Among their recommendations one was to merge the two departments. Thus in 1994, the current Department of Occupational Safety and Health was established (Paul Obua & Andrew Okimait, 2001).

Occupational Health in Uganda received fresh attention when a new national policy to protect workers'health came into force early in 2008.

Safety and Health Legislation *There are two important laws:*

Criminal law *imposes sanctions on behalf of the government in respect of a failure to comply with legal requirements*

Civil law *is the mechanism by which an injured person may seek compensation from another, for example compensation incurred from a work-related injury occurred.*

OHS Act; (Refer to Occupational Health and Safety ACT 2016)

Employee Rights under OHSA; Rights to-

- *A safe and healthy working environment*
- *Know about hazardous chemicals*
- *To trainings*
- *View hazardous exposure and medical records*
- *To participate in OHSA inspections*

Benefits of Good OSH

- ✓ *Reduced lost time*
- ✓ *Reduced need to provide cover for personnel absent from work because of injury*
- ✓ *Reduce sick pay payments*
- ✓ *Reduce need for time-consuming accident investigation and consequent saving in management time*
- ✓ *Reduced potential for prosecution and other enforcement action by authorities*
- ✓ *Reduction in the number of claims for compensation by injured personnel*
- ✓ *Improved morals*
- ✓ *Improved image of company, both in terms of employer and competent contractor*

DEFINITIONS

Consultation: *is a two-way exchange between employers and employees that involves:*

- ✓ *Sharing information about health and safety*
- ✓ *Giving employees opportunity to express their views*
- ✓ *Taking those views into account*.

Employee: *a person employed under a contract of employment or contracts of training.*

Hazard: *a source or situation with a potential to cause injury, illness or disease. Include Physical, Mechanical, chemical, Ergonomic, and psychosocial Hazards.*

Hazard identification: *the process of recognizing that a hazard exists and defining its characteristics.*

Health and Safety Representative *(elected to represent employees of a designated workgroup)*

Near miss: *potentially significant event that did not occur due to prevailing conditions, but could have resulted from a sequence of events that did occur.*

Risk: *the likelihood of an injury, illness or disease occurring*

Management commitment to OHS is demonstrated by

- ✓ *Allocating sufficient resources (financial and human)*
- ✓ *Establishing OHS structures to support employees*
- ✓ *Designating a management representative for overseeing the functionality of OHS*

Objectives of OHS Representatives:

- *Ensure a safe working environment without health risks.*
- *Implement and maintain safe systems of work; Consult with all employees on OHS matters*
- *Take reasonable steps to ensure risk are controlled at the client's workplace.*
- *Monitor and review the effectiveness of measures to protect employees.*
- *Ensure compliance with legislative requirements and current industry standards.*
- *Provide employees with necessary information, instruction, training and supervision*

RESPONSIBILITIES

Managers and Consultant *(they implement policies).*

- ✓ *Gathering information to determine:*
  - *The safety at each client site.*
  - *Tasks to be undertaken by the employee at workplace.*
  - *The qualifications and experience required by the employee to perform the work adequately and safely*
- ✓ *Evaluate, monitor and review the client's OHS systems and workplace to ensure the client is providing and maintaining to the employee a work environment that is safe and without risk to health*
- ✓ *Determine adequacy of risk control measures and if required negotiate change.*
- ✓ *Consult with employees in the development, promotion and implementation of health and safety policies and procedures.*
- ✓ *Ensuring that on-hired employees receive training in the safe performance of their assigned tasks by the client.*
- ✓ *The provision of resources to meet the health and safety commitment.*

A) Employees are to:

- ✓ *Report all identified hazards, near misses and injuries to your Consultant or to the appropriate client's representative when at a client's site*
- ✓ *Participate in skills development and training to actively improve competencies*
- ✓ *Follow all health and safety policies and procedures.*
- ✓ *Follow safe work procedures as set down by Workforce or the client.*
- ✓ *The consumption of alcohol or the use of illegal drugs whilst working is a dangerous practice and is not permitted.*

Employers should:

- *Carry out a risk assessment of the workplaces and work activities they are responsible for and through implementation of the risk assessment findings, remove or, as far as is reasonably practicable, reduce the risk of the identified hazards causing harm to workers*
- *Provide PPE where other means fail to reduce the risk presented by hazards in the workplace to an acceptable level;*
- *Carry out an assessment to identify the most appropriate types of PPE for use in the workplace.*

Where PPE is required it must:

- *Be appropriate for the risks and for the working environment;*
- *Take account of the user's health, ergonomic, physical and other factors adequately control the risk presented by the hazard without increasing overall risk experienced by the worker;*
- *Be supplied and replaced free of charge;*
- *Comply with relevant legislation implementing the European Directives concerning the design and manufacture;*
- *Where reasonably practicable provide a range of PPE to allow workers to choose the equipment that best suits their working environment and routines, but always ensure that the alternatives made available provide the level of protection required;*
- *Provide training, instruction or information, including refresher training and demonstrations, as appropriate in the use and care of the PPE provided;*
- *Ensure face fit testing is undertaken for close fitting respiratory protective equipment;*
- *Establish a system of recording the issue of PPE and of monitoring, examination and repair for PPE and to allow for the reporting of missing or lost items.*

OHS MANAGEMENT SYSTEM (OHSMS) / PROGRAM

The means by which an organisation controls risk through the management process. This includes the organisational structure, planning activities, responsibilities, practices, procedures, processes and resources for developing and achieving, reviewing and maintaining the organisation's OH&S policy.

It aims at providing a method to assess and improve performance in the prevention of workplace incidents and accidents via the effective management of hazards and risks in the workplace. It is a logical, stepwise method to decide what

needs to be done, how best to do it, monitor progress toward the established goals, evaluate how well it is done and identify areas for improvement

The purpose of an OHS Management System (also known as an OHSMS) is to systematically reduce the possibility of accident, illness, injury or fatality in the workplace by ensuring that hazards in the workplace are eliminated or controlled in a systematic manner

The enterprise's OSHMS is part of a global management system, which is based on the principles of ISO 9001 and also includes environmental and quality topics. Its main OSH related goal is the achievement of a zero rate of industrial accidents. In the beginning, the OSHMS was initialized by the 'Prevention and Protection committee', which consists of employees' and employers' representatives. After this, each department started to formulate points of focus in the effort to realize a safer and healthier environment.

Why Managing Health and Safety is important

- Moral and Ethical Issues Respect Human Resources. (Life/health of workers is very important).
- Legal Issues Claims for compensation. Disruption during the legal process.
- Financial Issues Increase of Insurance Premium Direct Lost
- Indirect Lost Benefits of Good OSH
  - *Reduced lost time*
  - *Reduced need to provide cover for personnel absent from work because of injury*
  - *Reduce sick pay payments*
  - *Reduce need for time-consuming accident investigation and consequent saving in management time and production down-time*
  - *Reduced potential for prosecution and other enforcement action by authorities*
  - *Reduction in the number of claims for compensation by injured personnel*
  - *Improved morals*
  - *Improved image of company, both in terms of employer and competent contractor*

Safety Programmes

Traditional approach *is the use of safety programmes which are focused on compliance with standards/regulations, they do not have strong or any feedback mechanisms. The performance of a programme is measured using (reactive) indicators such as the number of accidents, injuries etc.*

Four elements common to general systems theories are input, process, output and feedback. Systems are also classed as being either open or closed.

- **Open systems** *interacts with the external environment and is subject to external influences.*
- **Closed systems** *do not interact with the environment and so their ability to adapt or respond to changing internal conditions is limited.*

Key Elements of OHS Management System

Policy; *Effective health and safety policies set a clear direction (by setting goals) for the organisation to act upon. It demonstrates the formal commitment towards good OH&S by top management.*

- ✓ *They contribute to all aspects of business performance as part of a demonstrable commitment to continuous improvement.*
- ✓ *Responsibilities to people and the environment are met*
- ✓ *Stakeholders' expectations in the activity are satisfied.*
- ✓ *There are cost-effective approaches to preserving and developing physical and human resources, which reduce financial losses and liabilities.*

Organising; *an effective management structure and arrangements are in place for delivering the policy.*

- ✓ *All staff are motivated and empowered to work safely and to protect their long-term health, not simply to avoid accidents.*
- ✓ *There is a shared common understanding of the organisation's vision, values and beliefs.*
- ✓ *A positive health and safety culture is fostered by the visible and active leadership of senior managers*

Planning; *there is a planned and systematic approach to implementing the health and safety policy through an effective health and safety management system*.

Planning for OH&S involves:

- ✓ *Identifying requirements for the system - what needs to be done*
- ✓ *Setting clear performance criteria - what is to be done*
- ✓ *Identifying who is responsible - who gets it done*
- ✓ *Setting time scales - when it should be done by*
- ✓ *Identifying the desired outcome - what should be the result*

Measuring Performance; *Performance is measured against agreed standards to reveal when and where improvement is needed.*

The key purposes of measuring performance are to:

- ✓ *Determine whether OH&S plans have been implemented and objectives achieved;*
- ✓ *Check that risk control measures are in place and are effective;*
- ✓ *Learn from system failures such as areas of non-compliance, accidents and incidents.*
- ✓ *Promote the implementation of plans and risk controls by providing feedback to all parties*
- ✓ *Provide information that can be used to review, and if necessary to improve aspects of an OH&S management system.*

Organisation's performance measurement system incorporate both active and reactive monitoring data

Examples of active monitoring data are:

- ✓ The extent to which plans and objectives have been set and achieved.
- ✓ Whether a director for OH&S has been appointed.
- ✓ Whether a safety policy has been published.
- ✓ The numbers trained in OH&S.
- ✓ Number of risk assessments completed.
- ✓ Extent of compliance with risk controls.
- ✓ Extent of compliance with statutory requirements Frequency of OH&S audits.
- ✓ Frequency and effectiveness of OH&S committee meetings.
- ✓ OH&S specialist reports.
- ✓ Health Surveillance reports.
- ✓ Workplace exposure levels.
- ✓ Personal protective equipment use.

Reactive monitoring Systems

Should be used to investigate, analyse and record OH&S management system failures including:

- ✓ *Unsafe Acts*
- ✓ *Unsafe conditions*
- ✓ *Lost time accidents*
- ✓ *Major accidents & fatalities*
- ✓ *Sickness absences*
- ✓ *Criticisms made by regulatory agency staff*
- ✓ *Complaints made by members of the public.*

Auditing and Reviewing of Performance;

- ✓ *OH&S auditing is a process whereby an organisation can review and continuously evaluate the effectiveness of their OH&S management system.*
- ✓ *Safety auditing is a positive and proactive means of checking the safety performance of an organisation.*

Key Requirements of Auditing

Scheduling; *an annual plan should be prepared for carrying out internal safety audits.*

Management Support*; to be of value, senior management should be fully committed to the concept of auditing and its effective implementation within the organisation*.

Audits; *Audits provide a comprehensive and formal assessment of the organisation's compliance with OH&S procedures and practices. It should include a detailed written assessment of OH&S procedures, the levels of compliance with procedures and practices and should where necessary identify corrective actions.*

Auditors; *One or more persons may undertake audits. They should be independent of the part of the organisation or the activity that is to be audited.*

Audit results; *At the end of the audit, and before submitting their report, the auditor or the Audit team should hold a meeting with the responsible manager of the audited area. The main purpose of such a meeting is to communicate the results of the audit to the responsible manager in order to ensure that it is understood and agreement is reached.*

Content of the Report; *The content of the final audit report*

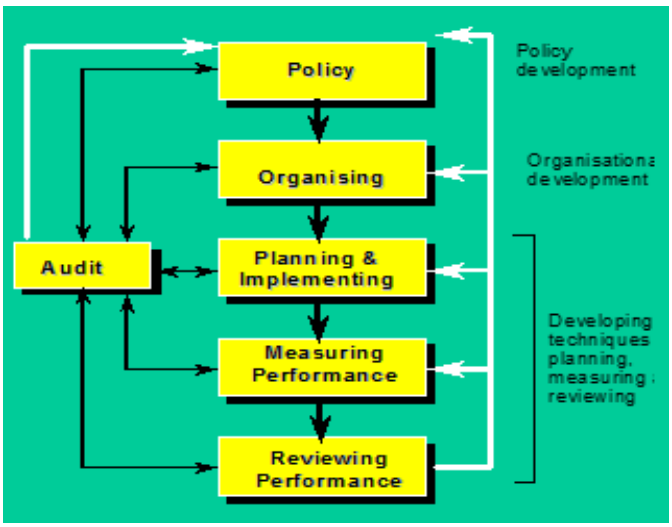
- ➢ *The audit objectives and scope*
- ➢ *The particulars of the audit plan, identification of the members of the auditing team and the audited representative, dates of audit and identification of the area subject to audit;*
- ➢ *The identification of reference handbooks used to conduct the audit;*

- *The cases of non-conformance.*
- *The auditors assessment of the degree of conformity to OHSAS 18001*
- *The ability of the OH&S management system to achieve the stated OH&S objectives*

Key requirements of Auditing; *Relevant documentation should be examined. This may include.*

- ✓ *OH&S management system documentation.*
- ✓ *OH&S policy statement.*
- ✓ *OH&S emergency procedures.*
- ✓ *Permit to work systems and procedures.*
- ✓ *Minutes of OH&S meetings.*
- ✓ *Accident/Incident reports and records.*
- ✓ *Training records.*
- ✓ *Reports or communication with the enforcing authority.*

HSE Model (HSG65)



The common problems encountered in the development of occupational health service include:

1. Lack of awareness among workers, employers, policy makers, health professionals and public at large.

2. Lack of trained human resource.

3. Inadequate, inaccessible, and inequitably distributed health service institutions.

4. Lack of multidisciplinary staff, absence of field-testing equipment for conducting environmental and biological monitoring of the work place and the health of the workers.

5. Insufficient budget for carrying out regular inspections, conducting research activities. 6. The characteristics of the workers, the majorities are poor, illiterate or poorly educated.

7. Poor working environment.

8. No specific regulation/ legislation on occupational health and safety issues.

Characteristics of a good OHSMS

- *Management Commitment*
- *Worker Involvement; Employees must be involved in the safety and health management system in at least three meaningful, constructive ways in addition to their right to report a hazard.*
- *Hazard Identification, including workplace inspections for violations of mandatory health and safety standards*
- *Hazard Prevention and Control*
- *Safety and Health Training*
- *Program Evaluation*

Health and safety benchmarking; *is a planned process by which an organization compares its health and safety process by which an organization compares its health and safety processes and performance with others to learn how to;*

- *Reduce accidents and ill-health*
- *Improve compliance with health and safety law*
- *Cut compliance costs*

The benchmarking process

Step 1; **Deciding what to bench mark;** *prioritizing high risk and high hazardous areas got from feedback from safety monitoring activities (risk assessment processes and accident data should identity these priorities). Consultation with the work force and trade union should take place at this stage.*

Step 2; Deciding where you are; *Identifying the current level of performance and desired improvement with reference to legal requirements. Current level can be measured by audit/questionnaire approach.*

Step 3; Selecting Partners; *for both internal and external benchmarking exercise to learn from other organizations (similar organizations, trade associations and local bench marking clubs).*

Step 4; working with your partner; *information is shared and compared, confidentiality should be respected by all parties with good understanding of partner's process activities and business objectives.*

Step 5; Acting on lessons learned; *Learn to know about the individual organization performance compared with the working partners and to take action to improve performance.*

According to Health and safety any action plan should be 'SMARTT' that is;

S *- Specific*

M *- Measurable*

A *- Agreed*

R *- Realistic*

T *- Trackable*

T *– Time bound*

Action plan *is should identify a series of recommendations and the people responsible to act on them with a specified time frame for implementation*.

To succeed in health and safety benchmarking; should have;

- ✓ *Management resource commitment*
- ✓ *Open and participatory approach to health and safety thus information sharing for internal and external organizations*
- ✓ *Employee involvement*
- ✓ *Meaningful data comparison*
- ✓ *Adequate research, planning and preparation*

The main advantages are that they encourage:

- ✓ *Systematic root analysis of hazards, risks and incidents;*
- ✓ *Strong awareness of hazards and risks;*
- ✓ *improved transparency concerning internal processes;*
- ✓ *Better communication among employees;*
- ✓ *Stronger motivation and identification of employees with their company;*
- ✓ *A more extensive and integrated point of view in the sense of the working environment;*
- ✓ *Better measuring of the occupational safety and health performance*

Barriers to the Desired Behavior

Environmental:

- ✓ *Poor Housekeeping*
- ✓ *Unsafe equipment and facility design*
- ✓ *Inadequately maintained equipment and facilities*
- ✓ *Rules, policies, and procedures that are unneeded or cumbersome*

Person:

- *Disagreement on safer practices and behaviors*
  - ✓ *Lack of self-worth – an unhealthy desire to be accepted*
  - ✓ *Lack of relationship and camaraderie*
  - ✓ *Lack of communication and interaction skills*
  - ✓ *Lack of personal responsibility (victim mentality)*
  - ✓ *Lack of coach-ability*
  - ✓ *Macho / hero mentality*
  - ✓ *Physical impairment*

Behavioral:

- ✓ *Few consequences for less-safe behavior*
- ✓ *Few consequences for safer behavior*
- ✓ *Emphasizing outcomes over process*
- ✓ *Condoning less safe behavior*
- ✓ *Management systems that are overly complex or unreliable*
- ✓ *Measurement and recognition systems that reward undesirable behaviors*

The advantages of an OHS management system include:

- *The ability to organize work properly, holding employees with OHS responsibility accountable for their decisions and actions;*
- *The development of management commitment to OHS, that can be viewed as a necessary condition for safe workplace*
- *The improvement of employees' participation in the OHS process which helps to encourage upward communication and enhance their involvement in decision making process*
- *The ability to improve employees' training through the comprehensive assessment of the major organizational needs;*
- *OHS management system affects the functioning of reward system that helps to reinforce the required OHS behaviors of employees, and at the same time, helps to improve their productivity and overall business performance;*
- *OHS management system improves communication through consultation (the proper consultation*

between managers and employees helps to improve a positive health and safety organizational culture);

- *OHS management system helps to identify hazards, assess and control risks, reducing the incidence, severity and direct and indirect costs of injuries/diseases in the workplace.*
- *meeting the so-called 'due diligence' requirements through implementation of a health and safety system that helps to demonstrate that managers effectively meet its legal responsibilities"*
- *The system can be properly audited and monitored in order to provide verification that the expectations are met, identifying and correcting any gaps in the system.*
- *the system helps to reduce the cost impact on the business, avoiding workplace injuries and illnesses reduction of the incidence of workers' compensation claims;*
- *minimizing of all types of work stoppages through the effectiveness of safety disputes;*
- *improvement of the workers' morale that leads to higher levels of production;*
- *Improvement of the existing working practices/or methods that leads to the clients' satisfaction*

Limitations of OSHMS

- *The production of documents and records needs to be controlled carefully to avoid defeating the purpose of the system by drowning it in excessive paperwork.*
- *Imbalances between management processes (quality, OSH, environment) must be avoided to prevent dilution of requirements and inequalities in focus.*
- *The lack of careful planning and full communication prior to the introduction of an OSHMS programme can raise suspicions about and resistance to the change.*
- *OSHMS usually puts greater emphasis on safety rather than health and with the risk of missing the onset of occupational diseases.*
- *Depending on the size of the organization the resources that are required to set up an OSHMS can be significant and should be the object of a realistic appraisal of overall costs in terms of implementation time, skills and human resources required to install and run the system*

Risk assessment and management

The essential purpose of OSH is the management of occupational risks. In order to do that, hazard and risk assessments have to be carried out to identify what could cause harm to workers as well as property so that appropriate preventive and protective measures can be developed and implemented.

Steps in risk assessment

STEP 1 *Identify the hazards*

STEP 2 *Decide who might be harmed and how*

STEP 3 *Evaluate the risks and decide on precautions*

STEP 4 *Record your findings and implement them*

STEP 5 *Review your assessment and update if necessary*

Key elements for a good OSH Management System

i. *Make a careful assessment of the organization's needs in relations to its means*
ii. *Adapt the OSHMS accordingly*
iii. *Ensure that the system stays focused on the performance of preventive and protective measures*
iv. *Keep in mind that it is designed to improve rather than to justify itself*
v. *Ensure that audits contribute to the continual improvement process rather than becoming a mechanism for improving audit scores only*
vi. *Remember that a management systems' approach is only as good as the OSH framework or programme in place in the organization;*
vii. *OSHMS programmes must function within the national OSH legislation framework and the organization must ensure that the system includes a review of regulatory requirements and is updated regularly to integrate them;*
viii. *OSH related training for the implementation of the OSHMS programme should be carried out on a continuous basis at all levels, from top managers to shop floor workers*
ix. *Communication channels between the different levels of the organization need for the system to focus on people.*
x. *Existence of effective social dialogue (direct involvement and consultation). Workers and their representatives should be given the opportunity to fully participate in the management of OSH in the organization*
xi. *A system is successful only when all the stakeholders are given responsibilities in running it.*
xii. *The future of OSHMS lies in striking the right balance between voluntary and mandatory approach*